

Crown Ridge (Mountain / Military / Command Networks & Strategic Positioning)

Executive Summary

This document is a technical specification and design brief for a command-network warfare, directed, coordinate, elevation control, and base-base coordination. The terrain features steep, dense ridges, forested/burned, command ridges, forested areas, elevated artillery platforms, and multi-tier mountain roads. The main installation (Observer Protocol) is autonomous command-network hub, sensor-rely chains, air-ground coordination, and logistics under high-altitude constraints. In AR/VR context, the main anchors at "4-5 m" view distance, coordinate a 10-15 m physical area (approximate "1 km" of mountainous terrain). Key features include command-rely chains, multi-tier defensive systems, elevation-driven observation, and elevated high-ground control.

Design Goals & Gameplay

Command/Network Warfare

Control ridge and/or command-network nodes, sensor hubs, and strategic communication lines. Destroying or capturing ridges affects control of entire sectors.

Elevation & Strategic Positioning

High ground provides strategic advantage, sensor coverage, and defensive lines. Controlling ridges is essential for long-range dominance.

Autonomy & Command-Chain Logic

Autonomous & sensor-rely chains from position command-network integrity, avoid exposed positions, and uses burners to break line-of-sight.

Sensor & Information Warfare

Use air, ground, and elevation gradients detect sensors. Relay towers on peaks restore clarity. Destroying ridges creates massive blind zones across the ridge.

Logistics & Resources

Mountain supply, command networks, and logistics serve as supply nodes. Supply convoys must navigate narrow roads and avoid ambush-point passes.

Deployment & Progress

A main objective is to capture the ridge, using autonomous sensors, command-rely chains, or mobile mountain infantry. Blue deploys from the southwest valley command. Red deploys from the northeast ridge cluster.

Spatial Constraints (Nreal AR)

Device Capabilities

Screen Size: 12" diagonal (20:9 aspect ratio), 1080x1800 per eye resolution. Optimal viewing distance: ~4-5 m.

Playable Area & Scaling

10-15 m physical area (~1 km² mountain zone (1-100 scale)). Designed for testing AR.

Tracking & Occlusion

Unity has tracking device horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view, assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000x1000 grid (1 unit = 1 m). Origin (0,0) = southeast corner.

Key Locations

Base 1 (SW Ridge): Command Base 1 (0,0). High ridge, 100m high. Main command center with radar, radio, and AR emitters.

Base 2 (NE Ridge): Command Base 2 (1000,1000). 140m high in forested high-ground cluster with radar tower and AR emplacements.

Central Valley Network: 1000x1000. 200m high. 200m high in forested high-ground cluster with radar tower and AR emplacements.

Command Ridge Network: 1000x1000. 100m high in ridge network. 100m high in ridge network. 100m high in ridge network.

Observer Ridge Network: 1000x1000. 100m high in ridge network. 100m high in ridge network. 100m high in ridge network.

Ridge Pass Corridor: 1000x1000. 100m high in ridge network. 100m high in ridge network. 100m high in ridge network.

Lower Valley Road: 1000x1000. 100m high in ridge network. 100m high in ridge network. 100m high in ridge network.

Peak A (SW Ridge Peak): 1000x1000. 100m high in ridge network. 100m high in ridge network. 100m high in ridge network.

Peak B (NE Ridge Peak): 1000x1000. 100m high in ridge network. 100m high in ridge network. 100m high in ridge network.

Neutral Forested Outcrop: 1000x1000. 100m high in ridge network. 100m high in ridge network. 100m high in ridge network.

Chokepoints & Sightlines

Ridge Pass Corridor

Extremely narrow, ideal for defensive artillery and command-rely protection.

Off-Road Burner Network

Interior routes and firing points, strong defensive positions.

Central Valley Road

Long range, sightlines, command-network dominance possible.

Lower Valley Road

Safe, but predictable, vulnerable to aerial recon.

Sightlines

Peak A provides long-range sensor coverage (~200m). Ridge peaks and elevation gradients create blind zones.

Deployment Zones

Base 1 (SW Ridge): 1000x1000. 100m high in ridge network. 100m high in ridge network. 100m high in ridge network.

Base 2 (NE Ridge): 1000x1000. 100m high in ridge network. 100m high in ridge network. 100m high in ridge network.

Red Outpost: 1000x1000. 100m high in ridge network. 100m high in ridge network. 100m high in ridge network.

Neutral Forested Zone: 1000x1000. 100m high in ridge network. 100m high in ridge network. 100m high in ridge network.

Assets & Environment

Terrain

Off-road, plateau, valleys, rock formations.

Vegetation

Some dense shrubs, pine clusters, grass patches.

Buildings & Props

Military: Bunkers, AR gun, radar dishes, helipads, command ridges. Industrial: Generators, pipelines, control towers, logistics. Trucks, crates, fuel drums, drums.

Effects

Fog, wind gusts, dust, smoke flares (optional), night-vision effects.

Audio

Wind, distant artillery, radio chatter, aircraft movement.

LOD & Budget

Asset Count

Asset Count: 1000. Total Size: 1000 MB.

Texture Budget

Texture Budget: 1000. Total Size: 1000 MB.

Model Budget

Model Budget: 1000. Total Size: 1000 MB.

Shader Budget

Shader Budget: 1000. Total Size: 1000 MB.

Sound Budget

Sound Budget: 1000. Total Size: 1000 MB.

Animation Budget

Animation Budget: 1000. Total Size: 1000 MB.

Scripting Budget

Scripting Budget: 1000. Total Size: 1000 MB.

AI Budget

AI Budget: 1000. Total Size: 1000 MB.

Physics Budget

Physics Budget: 1000. Total Size: 1000 MB.

Network Budget

Network Budget: 1000. Total Size: 1000 MB.

UI Budget

UI Budget: 1000. Total Size: 1000 MB.

Localization Budget

Localization Budget: 1000. Total Size: 1000 MB.

Testing Budget

Testing Budget: 1000. Total Size: 1000 MB.

Deployment Budget

Deployment Budget: 1000. Total Size: 1000 MB.

Documentation Budget

Documentation Budget: 1000. Total Size: 1000 MB.

Marketing Budget

Marketing Budget: 1000. Total Size: 1000 MB.

Legal Budget

Legal Budget: 1000. Total Size: 1000 MB.

Post-launch Budget

Post-launch Budget: 1000. Total Size: 1000 MB.

Overall Budget

Overall Budget: 1000. Total Size: 1000 MB.

Summary

Summary: 1000. Total Size: 1000 MB.

Conclusion

Conclusion: 1000. Total Size: 1000 MB.

Appendix

Appendix: 1000. Total Size: 1000 MB.

References

References: 1000. Total Size: 1000 MB.

Index

Index: 1000. Total Size: 1000 MB.

Glossary

Glossary: 1000. Total Size: 1000 MB.

Change Log

Change Log: 1000. Total Size: 1000 MB.

License

License: 1000. Total Size: 1000 MB.

Disclaimer

Disclaimer: 1000. Total Size: 1000 MB.

Privacy Policy

Privacy Policy: 1000. Total Size: 1000 MB.

Terms of Service

Terms of Service: 1000. Total Size: 1000 MB.

FAQ

FAQ: 1000. Total Size: 1000 MB.

Contact

Contact: 1000. Total Size: 1000 MB.

Support

Support: 1000. Total Size: 1000 MB.

Feedback

Feedback: 1000. Total Size: 1000 MB.

Partners

Partners: 1000. Total Size: 1000 MB.

Press

Press: 1000. Total Size: 1000 MB.

Media

Media: 1000. Total Size: 1000 MB.

Events

Events: 1000. Total Size: 1000 MB.

Jobs

Jobs: 1000. Total Size: 1000 MB.

Partnerships

Partnerships: 1000. Total Size: 1000 MB.

Investors

Investors: 1000. Total Size: 1000 MB.

Advisors

Advisors: 1000. Total Size: 1000 MB.

Board of Directors

Board of Directors: 1000. Total Size: 1000 MB.

Executive Team

Executive Team: 1000. Total Size: 1000 MB.

Senior Management

Senior Management: 1000. Total Size: 1000 MB.

Key Personnel

Key Personnel: 1000. Total Size: 1000 MB.

Organizational Chart

Organizational Chart: 1000. Total Size: 1000 MB.

Company History

Company History: 1000. Total Size: 1000 MB.

Future Plans

Future Plans: 1000. Total Size: 1000 MB.

Investment Opportunities

Investment Opportunities: 1000. Total Size: 1000 MB.

Partnership Opportunities

Partnership Opportunities: 1000. Total Size: 1000 MB.

Media Inquiries

Media Inquiries: 1000. Total Size: 1000 MB.

Press Releases

Press Releases: 1000. Total Size: 1000 MB.

Media Kit

Media Kit: 1000. Total Size: 1000 MB.

Media Contact

Media Contact: 1000. Total Size: 1000 MB.